

Introduction To JAVA

Why **JAVA**?: It solves some main problems of c, c++.

1] **SIMPLICITY**: JAVA c, c++ se ziyada simple thi.

2] **SECURITY**: JAVA c, c++ se ziyada secure thi. **JVM** me **sandbox** model hota ha jo code ko secured environment deta ha.

3] **PORTABILITY**: c or c++ platform dependent languages ha, inka jo code compile hota tha vo sirf kisi particular platform k lia hi hota tha.

EX: compiled code 10100101 bnaya ha platform A (Processor+OS) ke lia, to ye code platform B(Processor+OS) ke lia work nahi krega.

HOW DID JAVA SOLVE THIS PORTABILITY PROBLEM?

JAVA took the [concept of **BYTECODE**]

EXAMPLE:

C++ me hume dono platform k lia alag-2 compile krna pdta tha.

C++ code [Hello.cpp] => compiler => machine code [010.....] => platform A

C++ code [Hello.cpp] => compiler => machine code [110.....] => platform B

COMPILER AND INTERPRETER: Dono ka kaam same hi ha [Source code ko machine code me change krna].

Compiler: convert code all at once, fir sara converted code ek sath run krta ha. Example: **c, c++**

Interpreter: Convert code line by line, or har converted code ko sath ke sath run bhi krta ha. Example: **Python**.

JAVA is both language: Kuki ye Interpreter or Just In Time [JIT] Compiler dono ka use :krta ha.

JAVA code [Hello.java] => compiler => **BYTECODE** [Hello.class] => **JVM** => **INTERPRETER + [JIT]**
COMPILER => machine code [110.....] => Platform A,B,C,D.....